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# AI-enabled prediction of sim racing performance using telemetry data

Fazilat Hojaji [a,*], Adam J. Toth [a,b,c], John M. Joyce [a], Mark J. Campbell [a,b,c]

[a] Esports Science Research Lab, Lero, The Science Foundation Ireland Centre for Software Research, University of Limerick, Limerick, Ireland
[b] Department of Physical Education & Sport Science, University of Limerick, Limerick, Ireland
[c] Sport and Human Performance Research Centre, University of Limerick, Ireland

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ABSTRACT

Despite the emerging and rapid progress of esports, approaches for ensuring high-quality analytics and training among professional and amateur esports teams are lacking. In this paper, we demonstrate how the application of data science techniques and Machine Learning (ML) approaches in esports, particularly in sim racing science, can illuminate the most important in-game metrics that dictate performance. Thus, using a professional racing simulator and MoTec i2 Pro (v1.1.5, Australia), we gathered extensive telemetry data from 174 participants, who completed 1327 laps on the Brands-Hatch circuit in the Assetto Corsa Competizione (v1.9, KUNOS Simulazioni). We clustered the obtained laps based on performance (lap-time), and then identified driving behaviors within performance groups. We also analyzed the feature subset obtained from a hybrid feature selection approach using two correlation analyses and three ML models.

The best model achieved a prediction accuracy of 97.19%, demonstrating that the model effectively captured the critical factors that influenced driving performance during a lap. The results confirm that *average speed* is the most important metric, followed by *lateral acceleration*, *steering angle*, and *lane deviation*. Our analyses offer key metrics for refining training tools and techniques in sim racing performance improvement.

## 1. Introduction

### 1.1. Background

The term *esports* refers to a form of organized, competitive video gaming. It involves individuals or teams competing against each other in video game tournaments, much like traditional sports. Esports has grown significantly in popularity over the years and is now a major industry with professional players, teams, leagues, and a substantial fanbase (Kovács & Szabó, 2022). According to the most recent analysis from the Consumer Technology Association (Sinclair, 2021), 519 million people will be watching esports worldwide in 2024 (Sinclair, 2021). Statista (Statista, 2023) predicted that the esports market will be valued at 1.87 billion US dollars in 2025, growing at a pace of 3.8%. As the popularity of esports has surged, there has been a growing demand for the analysis of performance strategies and the prediction of player behaviour, leading to the emergence of the new discipline of esports analytics (Schubert et al., 2016).

Esports analytics refers to the method of analysing esports-related data, particularly behavioural telemetry, to uncover meaningful patterns and trends among such data, and the use of visualization approaches to understand how these patterns are transmitted to support decision-making processes (Schubert et al., 2016). Esports analytics may be considered a part of game analytics or sports analytics (Drachen et al., 2013; Lindsey, 1959). Esports analytics have benefited from a variety of community-based projects aimed at helping players access, analyze, and derive meaning from data pertaining to their video game performance (Chu et al., 2019). Regardless, analytics have become a substantial element of the esports landscape. Nowadays, the majority of professional teams hire analysts to study their competitors, use data-driven approaches to identify their patterns and strategies, and devise counter-strategies, similar to the roles played by analysts in traditional sports (Schubert et al., 2016).

### 1.2. Related work

Within data science, artificial intelligence (AI) has plays a crucial role in leveraging advanced algorithms and computational techniques for data analysis and sports performance prediction (García-Aliaga et al., 2021),(Mittal et al., 2021). Machine learning (ML) is a branch of AI

* Corresponding author.
*E-mail addresses:* Fazilat.Hojaji@ul.ie (F. Hojaji), Adam.Toth@ul.ie (A.J. Toth), John.M.Joyce@ul.ie (J.M. Joyce), Mark.Campbell@ul.ie (M.J. Campbell).

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containing techniques to make computers learn from data (El Naqa & Murphy, 2015, pp. 3–11). The benefit of AI is that it can rapidly process large volumes of data, and data analysis techniques are continually evolving, enabling users to gain crucial information that is challenging to obtain manually (Li & Xu, 2021). To date, many researchers have made great efforts to apply statistical techniques, and computational tools in all aspects of sports (Beal et al., 2019; Li & Xu, 2021; Nadikattu, 2020). Artificial neural networks (Aggarwal, 2018), decision tree classifiers (Pang-Ning Tan, 2006), Markov processes (Ross, 2019), and support vector machines (Cristianini, 2000) are now being used in sports including basketball, soccer, and volleyball to predict performance (Claudino et al., 2019). In contrast to traditional sports, scientific methods and ML techniques have not been extensively used in the context of esports in general or in simulated racing (sim racing) in particular. The application of data science techniques in sim racing, which is relevant to the research reported in this work, could lead to technological enhancement in the computer-based simulator and could contribute to the direct improvement of the team and sim racer performance (de Frutos & Castro, 2021).

The majority of experience and research in this field to date has concentrated on psychological factors in sim driving. Starting with the study operating on simulated driving, Touryan et al. (Touryan et al., 2016) considered clustering participant behaviour to identify a number of both neural and eye-movement components, across different drivers. The authors used these component clusters to quantify the relative influence of common versus participant-specific features in the models of driving behaviour. Zhang et al. (Zhang et al., 2020) used a similar approach to target driver fatigue detection by clustering brain networks to extract dynamic features over longer time frames. The two-dimensional fatigue pattern was also recognized by the authors using PCNN-based image processing (PCNN- Pulse-Coupled Neural Network that is used to perform image analysis and processing tasks). In the case of classification techniques, existing studies are scarce. From this limited selection, prior studies have employed classifiers such as neural network (NN) (McCulloch & Pitts, 1943) or Support Vector Machine (SVM) (Cortes & Vapnik, 1995) for classification, feature extraction (Remonda et al., 2021) and pattern recognition (Von Schleinitz et al., 2019). As an example, Tjolleng et al. (Tjolleng et al., 2017) applied an artificial neural network model on ECG signals to classify driver's cognitive workload levels. ECG data were captured while participants performed a simulated driving task as the primary task with/without a working memory task (N-back) which acted as the secondary task. While there are a few studies dealing with the prediction of a driver performance in simulated driving in general (Bugeja et al., 2017; Gotardi et al., 2019; (de Winter et al.,); Remonda et al., 2021; Von Schleinitz et al., 2019), extremely little research exists to date exploring what constitutes optimal performance within competitive sim racing.

The lack of ML techniques and subsequent analysis in sim racing is unexpected, given the availability of data to the general public has significantly increased thanks to APIs and tools (prominent examples of which are; vTelemetry PRO (Renovatio, 2017) and Motec i2 Pro (Ltd, 2023) that grant access to telemetry data from sim racing video games. The abundance of telemetry data produced by sim telemetry tools enables the execution of fundamental analysis via ML techniques. Insights from telemetry data lead to a better understanding of the corresponding strengths and weaknesses of the driving behaviour and can foster performance improvements through more accurate tuning of their car setup as well as informing them on driving strategies and techniques (UNAmedia, 2023).

The only paper dealing with sim telemetry data is by Remonda et al., (Remonda et al., 2021), with the goal of predicting the driving performance and informing an autonomous system. While Remonda et al.'s research primarily focuses on a limited set of vehicle control behaviour, such as the steering wheel, throttle, pedal, and brake pedal, our work takes a distinct approach. Our research focuses on extracting key performance metrics and entails a thorough investigation of real-time driving data that extend beyond traditional vehicle control behaviour. An objective assessment of the characteristics of the driver's behaviour is obtained through a set of defined lap-based metrics derived from telemetry data. Our results not only validate but also extend the findings of Remonda et al. (2021) by presenting a detailed catalog of in-game metrics tailored to sim racing. We discuss how these metrics can be effectively utilized to evaluate and compare driver performance.

### 1.3. Research purpose

By leveraging the immense amount of telemetry data produced by sim telemetry tools and utilizing data science techniques, the purpose of this study is to model and analyze race driver behaviour and determine the most important in-game metrics that dictate performance for sim racing. Specifically, our study has three primary aims:

A1. To identify the metrics that best predict sim racing performance, across different levels of expertise.

A2. To utilize the identified important metrics for the investigation and visualization of behavioural patterns among racers.

A3. To determine the level of difficulty for each corner of the racing track through by conducting track segment analyses.

This paper is a significant extension of our previous works (Hojaji et al., 2022, 2023). The extension comprises (i) a fully generic set of car control metrics, (ii) a revised driving pattern investigation, (iii) an up-to-date performance metrics for lap and segments, (iv) a comprehensive review of machine learning and related work.

To the best of our knowledge, this is the first study to explore the application of ML techniques that carries out a comprehensive analysis of sim racing telemetry data to establish a robust measure of various aspects of sim racing performance.

## 2. Materials and methods

This section outlines the essential components of the study's methodology, including details regarding participants involved, the equipment utilized, and the preprocessing steps undertaken.

### 2.1. Participants

A total of 174 young, healthy adults (mean age 30.44 $\pm$ 10.19 years) were recruited from the University of Limerick student and staff population as well as simulated racing societies around the Republic of Ireland. Participants attended a testing session in the Esports Science Research Lab located in Lero, the SFI Research Centre for Software. Upon arrival, they were asked to fill out a general questionnaire to provide information on their gaming habits, real life driving experience, and basic demographic information. General questions were posed about driving passenger cars. The percentage of participants with a driving license was 85%.

### 2.2. Simulator environment and equipment

Participants raced on a custom racing simulator which included a Sparco Grid Q Fibreglass seat mounted on a SRC Pro V2 chassis. A high-end consumer-grade gaming PC was used with a single, large 32″ MSI MPG321QRF-QD widescreen monitor. The vehicle's control interface in the simulator consisted of a steering wheel (Logitech Pro and control pedals (brake, clutch, and accelerator (Logitech Pro). Fig. 1 displays the hardware components of the racing simulator. We utilized Assetto Corsa Competizione software (ACC; developed by Kunos Simulazioni, Rome, Italy), an advanced racing simulator which is designed to run on consumer-grade PCs. ACC already has a large number of precise car and track models. The Brands-Hatch Grand Prix circuit was chosen, and the car used was the McLaren 720s GT3. The official track divisions (named lap segment) depicted in Fig. 2 were used in the analysis. We also augmented ACC with MoTec i2 Pro, a professional standard telemetry


**Fig. 1.** Sim racing setup used by participants. The Logitech G Pro wheelbase and pedals were incorporated into the SRC Pro V2 Chassis, as well as the Sparco Grid Q Fibreglass seat.

software developed by MoTec Pty Ltd, and routed all telemetric data to the data analysis package.

### 2.3. Data collection and pre-processing

After completing the initial questionnaire, participants were instructed to sit at the racing station (Fig. 1) and to drive as quickly as they could while keeping the car on the track during the driving task. Participants completed X laps, and the total racing session lasted approximately 30 min.

The vehicle telemetry was recorded for all participants. To extract telemetry data from MoTec log files, we used MoTec i2 Pro (V1.1.5). Following the guideline to setup the ACC workspace on MoTec, we configured the software and defined particular settings (see supplementary file 1 to find MoTec and ACC setup) for track segment time, lap and driver dataset, and inbuilt maths and functions. In addition, we incorporated additional behavioural data features in MoTec by formulating specific features in telemetry data prior to starting data collection. The formulas used to calculate these additional features can be found in the supplementary file 8. Performance encompasses a variety of factors, including ideal tire wear, optimal wear of a vehicle's critical components, and optimal fuel efficiency (Remonda et al., 2021). In this work, driving performance was assessed via lap-time as well as the sector times.

We used Python (3.9) on the Anaconda 3 (Spyder 5.2) platform to implement pre-processing and analysis steps. We also used sklearn library for data standardization. In order to enhance transparency and reproducibility, all data processing scripts used in this study are publicly available in the Git repository.[1]

Fig. 3 provides a graphical overview of the experimental steps used for pre-processing and data analysis. We have provided a brief description of the pre-processing steps below.

We exported lap data containing general descriptions of the racing session (e.g., venue, track name, vehicle, duration) and lap-times and segment times for both straights and corners. We also extracted all data statistics corresponding to driver and vehicle in-game behaviour. Note that MoTec can acquire up to 74 data features including the driver's input data, chassis and suspensions data, brakes and tyres data, engine data, wheels data, balancing data, g-force and car's position. To capture the driver's behaviour during driving session, time-series data with a sample rate of 50 Hz were extracted. Therefore, in our experiment we have four datasets: lap-time data, lap-specific behavioural data, segment-specific behavioural data, and time-series driving data. The specification of these datasets is presented in Table A1 in the appendix. Examples of these data are available in the supplementary data files 2, 3, 4, and 5.

Pre-processing was conducted on the aforementioned datasets in

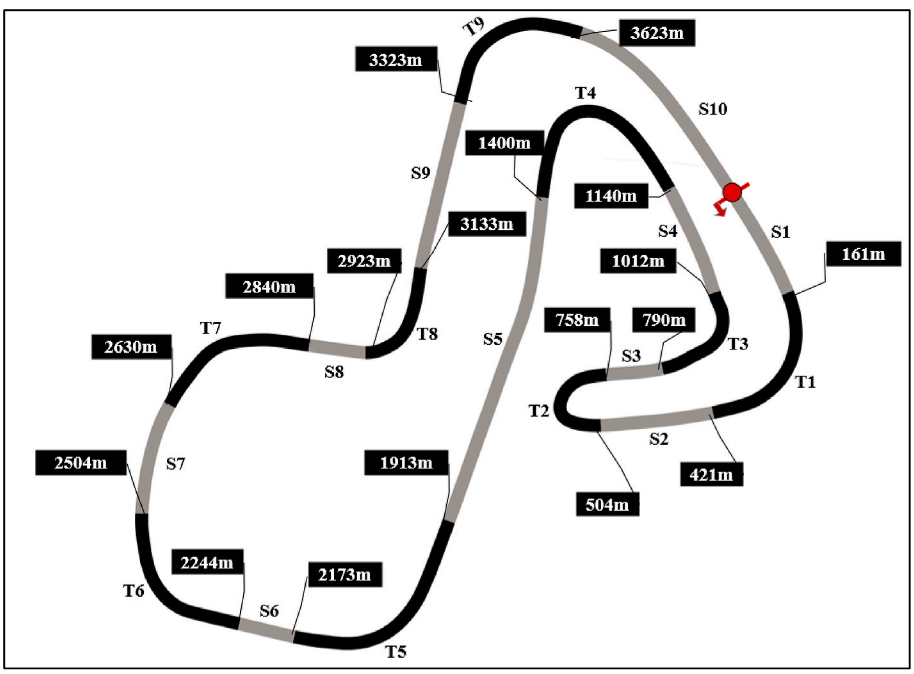


**Fig. 2.** The Brands-Hatch circuit. The figure shows the track division of the circuit complete with demarcated turns (T1 … T9) in black and straights (S1…S10) in grey colour. Track distances are indicated in meters.

[1] https://github.com/ESRLAccount/SimRacing_KPITelemetryDataAnalysis.

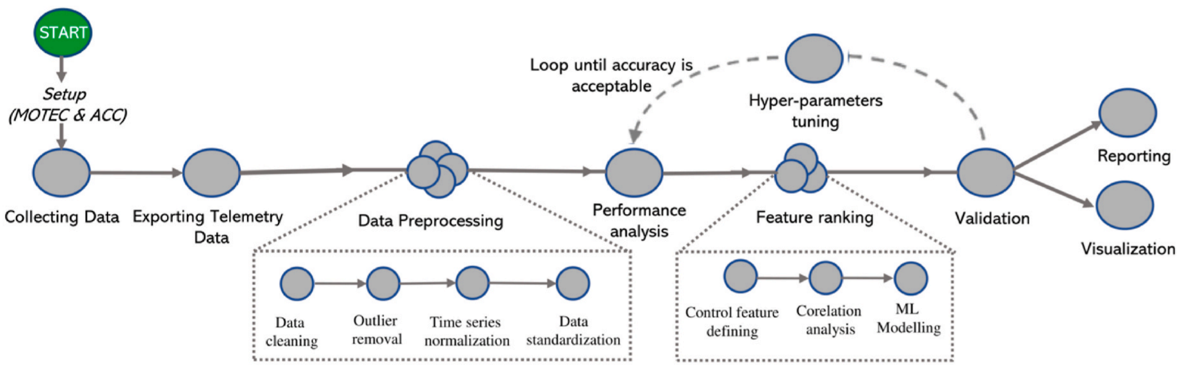


**Fig. 3.** Data collection, preprocessing and processing steps applied in the study.

accordance with the guidelines provided by Tabachnick and Fidell (Tabachnick et al., 2007). For the lap-specific behavioural and segment-specific behavioural datasets, missing data for certain features were replaced with the respective mean values for those specific data features. In the lap-time dataset, 12 invalid laps including laps with zero lap-time caused by a MoTec disconnection were omitted from the analysis. We also excluded *out laps*, the first lap taken after the driver exits the pits—and *in laps*—the laps taken after the driver returns to the pits as both are usually slow laps and not representative of driver performance. A total of 1327 laps were included and subjected to additional criteria for outlier removal using the z-score normalization method (Andrew and Siegel, 1998) and those laps were temporally isolated. We observed a marginal enhancement in accuracy when using a z-score threshold of 2.5 as opposed to evaluating at the z-score thresholds of 1 and 2, resulting in the exclusion of 32 laps. Additionally, we applied a custom normalization method to the time-series driving dataset. Because the number of rows for each lap varies depending on the length of the laps (i.e., a shorter lap has fewer time-series rows of data in telemetry data), we constructed one row per lap distance indicator by averaging all the data related to that lap distance. Additionally, we incorporated segment time from lap-time dataset into segment-specific behavioural dataset. Finally, as features in datasets have different scales (e.g., one feature in meters and another in degree), all datasets were standardized. Standardization, in the context of data preprocessing, refers to the transformation of data to have a mean of zero and a standard deviation of one (James et al., 2013). It is a crucial step in data preprocessing, especially when dealing with machine learning algorithms that perform better when features are on a similar scale. The same approach was applied across all segment-specific behavioural data for each participant.

As a consequence, two datasets were obtained, each containing 62 data features, one for analysing performance throughout the course of a lap and the other for analysing performance over a specific segment of the track. The lap-specific behavioural dataset comprises a total of 1327 laps for analysing performance throughout each lap, while the segment-specific behavioural dataset includes 15,694 rows for analysing performance over specific segments of the track. Both datasets are accessible in supplementary data files 6 and 7. Table 1 summarizes the statistics per lap and segment.

## 3. Analysis

This section delves into the detailed analysis of the collected data to achieve three primary objectives of the study. In our methodology (see Fig. 3), we utilized two datasets derived from the preprocessing step. The first dataset consisted of 62 features and 1327 rows, while the second contained the same number of features but with 15,694 rows. Subsequently, we applied clustering for performance analysis. The clustering results were then employed as a target for model training in feature ranking. Finally, we visualized and reported the results after validation. We have outlined the analysis steps in brief detail below.

### 3.1. Performance analysis

Throughout the methodology employed in this study (see Fig. 3), performance analysis is critical as it lays the groundwork necessary to address the three aims. To do so, we attempted to categorize the laps into different performance groups. We exclusively focused on lap-time as the primary performance metric, given that all drivers participated in the same track, utilized the same vehicle setup, and competed with the same game setup. We used both lap-specific behavioural and segment-specific behavioural datasets resulting from the pre-processing step for analysing performance throughout the course of a lap and for analysing performance over a specific segment of the track. The clustering algorithm was applied separately to each dataset. For the lap-specific behavioural dataset, the algorithm was applied to a one-dimensional array of lap times to categorize the data into different groups. Conversely, for the segment-specific behavioural dataset, the algorithm categorized different groups into segments using a two-dimensional array of lap time and segment time. We analyzed two K-value selection methods, namely Elbow method (Jain & Dubes, 1988), and Silhouette Coefficient (McCallum et al., 2000) to determine the optimal number of clusters for the dataset. We used the Python Scikit-Learn module for the implementation of Elbow, Silhouette score and the K-means algorithm (MacQueen, 1967). In this analysis, we specified hyperparameter settings, including *kmeans++* (Arthur & Vassilvitskii, 2007) for initialization, a value of 2 for n_init, a maximum of 1000 iterations for max_iter, and a random state of 42.

After selecting the optimal number of clusters using the K-value with the best performance, we then used the K-means method, the most commonly used clustering algorithm in many fields such as sport science (e.g., (Odierna & Silveira, 2020), (Remonda et al., 2021)), medical science (Shweta Jain and Jain, 2017) and health (Abdullah et al., 2022). The K-means algorithm selects the number of clusters (k) and initializes each cluster centroid in a different location within the dataset. In this method, the initial cluster centre (centroid) positions are selected randomly, then, centroids iteratively move and begin clustering the data based on the Euclidian distance between the data and the cluster's mean until no further movements are required and the clusters are determined (Maheshwari, 2020).

### 3.2. Feature ranking

#### 3.2.1. Vehicle control behaviour

In line with the first aim of the study (A1), which sought to identify the primary indicators of sim racing performance, we utilized both lap and segment-specific behavioural datasets obtained from performance analysis step that included performance group along with telemetry data relating to the driver's car control, such as speed, RPMS, steering angle, ROTY, brake, throttle, lateral and longitudinal acceleration.

**Table 1**
Overall performance statistics for laps and track segments, with average (mean), max (worst time), min (best time), and standard deviation (std). Brands-Hatch track consists of 19 segments including 10 straights (i.e., S1… S10) and nine turns (i.e., T1 … T9). All values represent time in seconds.

| | Lap | S1 | T1 | S2 | T2 | S3 | T3 | S4 | T4 | S5 | T5 | S6 | T6 | S7 | T7 | S8 | T8 | S9 | T9 | S10 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| mean | 93.39 | 4.0 | 4.1 | 4.3 | 6.5 | 2.2 | 4.9 | 5.3 | 5.3 | 12.0 | 3.7 | 3.6 | 4.3 | 4.6 | 4.1 | 2.1 | 4.8 | 6.0 | 4.0 | 7.4 |
| max | 137.54 | 16.8 | 8.4 | 18.1 | 20.2 | 4.3 | 17.0 | 28.6 | 21.0 | 30.6 | 5.9 | 12.6 | 14.3 | 20.1 | 10.9 | 20.2 | 14.4 | 17.3 | 23.0 | 22.2 |
| min | 85.05 | 3.5 | 3.5 | 3.7 | 6.0 | 2.0 | 4.4 | 4.8 | 4.8 | 11.2 | 3.0 | 3.3 | 3.7 | 4.2 | 3.2 | 1.7 | 4.0 | 5.6 | 3.1 | 6.9 |
| std | 7.90 | 0.8 | 0.5 | 1.2 | 0.8 | 0.2 | 0.7 | 1.5 | 1.0 | 1.6 | 0.4 | 0.7 | 0.6 | 0.9 | 0.8 | 1.3 | 0.8 | 0.6 | 1.1 | 0.8 |

Furthermore, to gain deeper insights from the data, we performed an analysis on time-series driver dataset across several features including throttle, brake, and steering angle, aiming to explore their characteristics and establish new aggregated metrics from telemetry data. Specifically, we derived three additional features from brake data, encompassing start, peak, and end positions, alongside the peak value for each brake in every lap. These features were obtained through the analysis of each time-series dataset. Employing the scipy Python library, we identified peaks in the brake signals, excluded smaller peaks, and determined the width of each peak. Subsequently, we averaged these features for each peak across all brake data. The same approach has been applied for calculating new aggregated metrics derived from steering angle (e.g., maximum value of steering, number of zero crossing, angular travel) and throttle (e.g., throttle application). In addition, we used Exploratory Data Analysis (EDA) (Tukey, 1977) to understand the distribution, range, variance, and statistical properties (i.e., max, min, mean, std) of each data column within datasets. To assist the reader with telemetry datasets, we present a glossary with the most important terms in Table A2 in appendix.

#### 3.2.2. Driving pattern investigation

Driving pattern investigation involves various techniques and methods to analyze and interpret driving behaviour, vehicle movement, and related data. Our second aim (A2) was to visualize the behavioural patterns of sim racers across an entire lap according to the most important metrics identified from the feature ranking. To achieve this, we used two different approaches, outlined below, for understanding, quantifying, and comparing patterns in driving data.

**Time-series Analysis**: analysing temporal aspects of driving data to understand patterns over time (speed variations, acceleration trends, etc.).

**Visualization Techniques**: utilizing graphs, charts, and heatmaps to visually represent driving patterns and trends.

#### 3.2.3. Feature selection

Feature selection is the process of creating a subset from an initial feature set using ML algorithms, which removes the redundant and irrelevant features and picks the relevant features of the dataset (Cai et al., 2018). There are three main categories of feature selection methods: wrapper methods, filter methods, and embedded methods. Researchers can also employ combinations of these techniques. The filter method selects the primary features simply based on the dataset features, without any classification model. The least significant features are rejected, while important features are chosen to predict outcome variable based on their statistical significance. In contrast, a wrapper technique selects the best subset from the feature set based on a particular ML algorithm. Embedded methods only perform feature selection during model training. It facilitates finding the optimal subset of features for that model (Ghosh et al., 2019). Hybrid methods have the potential for a better selection process. The main benefit of hybrid approaches is that they combine the best aspects of each feature selection technique, which raises prediction accuracy and makes computations simpler than with wrapper techniques.

Inspired by the approach proposed by Mandal et al. (Mandal et al., 2021), we applied a hybrid approach for feature ranking based on correlation analysis and ML algorithms. For feature ranking, we opted to utilize the mean values of each feature instead of incorporating the entire dataset comprising mean, minimum, maximum, and standard deviation. This decision was made to streamline the feature set, reducing dimensionality and potential noise in the data. Focusing solely on the mean values allows for a more concise representation of the central tendency of each feature, contributing to a more efficient and interpretable classification model. Initially, a Pearson correlation matrix (Witte & Witte, 2017) was created among all metrics to discard highly correlated ones. On inspection of the correlation matrix, we realized that many metrics obtained from telemetry data were highly correlated. The matrix also showed some interesting conclusions about the presence of multicollinearity, as there were multiple instances with absolute correlation coefficients >0.7. By multicollinearity, we mean collinearity between three or more metrics. In order to deal with multicollinearity, we used the Variance Inflation Factor (VIF) (James et al., 2013) to choose a subset of significant metrics, which is the most common technique used in detecting multicollinearity. In particular, we adopted an iterative procedure to compute VIF in multiple iterations recursively. At each iteration, the VIF for each metric is computed, and the metrics with the highest VIF value (>10) is considered a potential candidate for elimination. Once the metric has been removed, the process is then repeated by recalculating VIF for all remaining metrics. If there is no VIF value larger than 10, the procedure is terminated. Out of the 46 metrics that resulted from Pearson analysis, 21 were eliminated from the regression analysis because of multicollinearity. The remaining 25 metrics were input into ML incorporating. In this step, we conducted a bootstrapping analysis among various supervised machine learning algorithms. We used the most common and modern supervised machine learning techniques for classification, which are listed below, to build the prediction models. The models we used are Gradient Boosting (Chen & Guestrin, 2016; Friedman, 2001), Support Vector Machine (SVM) (Cristianini, 2000), and Random Forest (RF) (Breiman, 2001). These models were used to determine in-game metrics best predicted the performance of a sim racer.

For the implementation of these models, we utilized the scikit-learn Python library, which provides various tools for data preprocessing, model selection, and evaluation. Specifically, we utilized XGBoost (Chen & Guestrin, 2016), SVM, and RandomForestClassifier for building and training the models in Python.

Following the assessment of model performance, XGBoost was selected as the classifier for the extracted features. More details about the assessment are provided in Section 4.

*3.2.4. Track corner difficulty analysis*

Addressing the third aim (A3), which strived to determine level of difficulty associated with corners along the Brands-Hatch circuit, we attempted to classify lap segments into groups or clusters in such a way that items within the same cluster are more like each other than they are to those in other clusters. To do so, the elbow approach and the silhouette method along with the K-mean algorithm were applied to the segment-specific behavioural dataset obtained from the pre-processing step to cluster nine different corners. From the dataset, we employed the clustering algorithm on a subset of data pertaining to corners, comprising 7847 rows. In order to achieve more accurate results, we incorporated both segment time and segment length together with the top 10 important metrics identified from key performance metrics analysis (A1) for clustering corner segments.

## 4. Results

In this section, we present summary of the analyses conducted, encompassing performance analysis, KPI analysis, and track difficulty analysis. This section provides detailed insights into the outcomes of each analysis, explaining key findings and observations.

*4.1. Racing performance level*

Analysis via the elbow and silhouette methods indicated that the preferred number of clusters for lap-time performance is two or four. As we aimed to determine the main difference between elite and low skill sim races, we opted for a solution with two clusters using K-means.

Fig. 4 presents the results of clustering of laps. The cluster names refer to the lap-time, i.e., the SLOW refers to slower lap-times and FAST refers to faster lap-times. These clusters achieved a silhouette score of 0.81, indicating strong separation between clusters.

The FAST group exhibited a mean value of 96.63 ± 7.71(SD), with

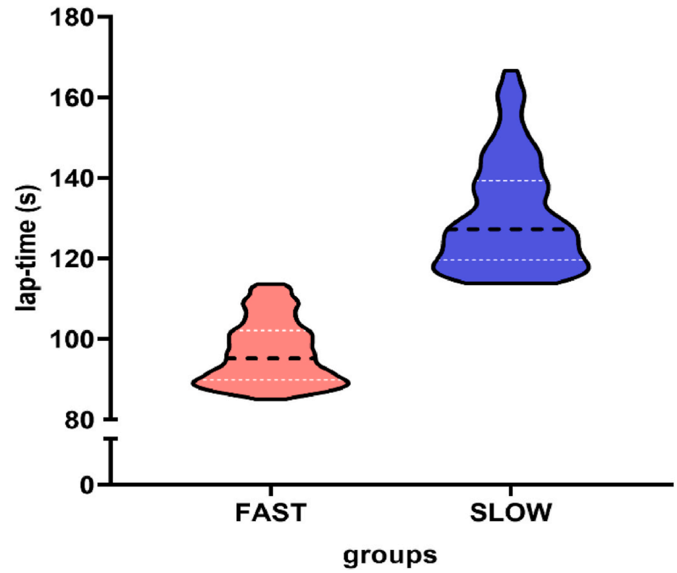


**Fig. 4.** Clustering visualization showcasing violin plots displaying the means and distributions, within each cluster. The thick line in the centre of each plot represents the median, while the two white lines represent the interquartile range. Wider areas of the violin plot represent a higher density of laps in the cluster for the given value; whilst a lower density is represented by smaller portions.

minimum of 85.04 and a maximum of 113.65. The SLOW group showed a mean value of 130.68 ± 13.3(SD), with a minimum of 113.77 and a maximum of 166.63 s. Furthermore, apart from categorizing the laps, we employed clustering methods to partition the segment-specific behavioural dataset. Consequently, we executed the same clustering technique on each segment independently to categorize the data within each segment. In total, we acquired 20 datasets of clustering outcomes, comprising one dataset for lap clustering results and 19 individual datasets, each corresponding to the clustering outcomes of a specific segment.

*4.2. Key performance metrics in sim racing*

Addressing the initial objective outlined in this study to identify the metrics that best predict sim racing performance (A1), we applied the datasets resulting from our performance analysis to a hybrid feature selection approach with the combination of two kinds of correlation analysis and classification techniques. With the data vector obtained from the correlation analysis, we applied three classification methods and recorded their corresponding accuracy. We divided the data into a training set (70%) and validation set (30%) in order to train the model. The model was trained using the training sets, and the accuracy of the predictions was assessed using the validation sets. To build more robust models, a 10-fold cross-validation procedure was utilized to evaluate our classifiers, and their precision score was calculated using mean absolute error, accuracy, F1 score, and area under curve (AUC). Fig. 5 illustrates a presentation of AUC's of the three classifiers. We found that the XGBoost model delivered the highest precision score among three algorithms. The AUC of the model was 0.971, the accuracy was 0.96, the F1-score was 0.93, and the absolute mean error was 0.011. So, we relied on the results of this algorithm for finding key performance metrics in sim racing. In addition, all algorithms produced better accuracy across all included ranks compared to the results of individual rank groupings, which is reasonable given that more data yields more accurate classification outcomes.

We used the Python shap package to explain and display the XGboost results. The utilisation of SHAP values is based on cooperative game theory and is used to make machine learning models more transparent

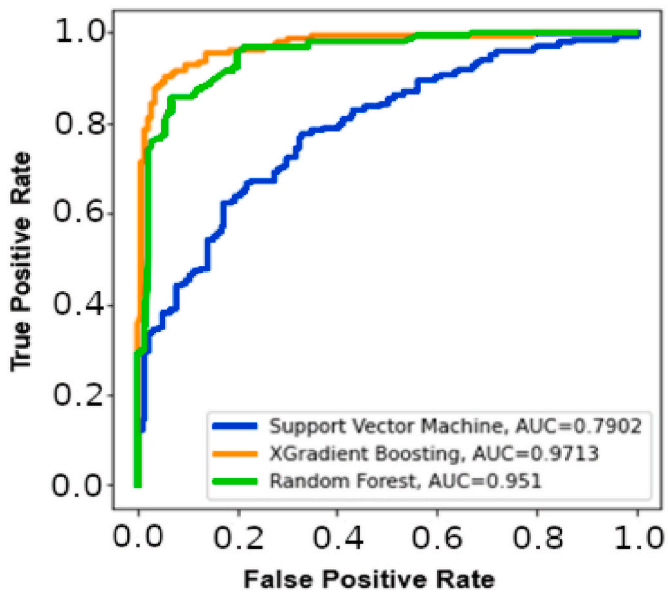


**Fig. 5.** AUCs representing different levels of performance quality for three classification algorithms. The lines closer to the diagonal, such as the SVM line (blue colour), are indicative of performance that may be considered suboptimal, while the curves away from the diagonal represent higher levels of performance. The closer the area is to 1, the better the model. Among three algorithms, XGBoost model has outperformed with the highest accuracy score.

and understandable (Lundberg & Lee, 2017). As a measure of the overall significance of each metric, machine learning models employ their coefficients. However, because these coefficients are scaled with the size of the variable, this may cause distortions and misinterpretations. Additionally, the feature's local importance and how it varies with lower and higher levels are not taken into account by the coefficient. SHAP shows the individual contribution or the importance of each metric relative to the importance of all other metrics on the prediction of lap-time, and this is why SHAP is useful for interpretability of models. The SHAP graphs shown in Fig. 6 was obtained by analysing the mean value of the absolute SHAP values of the top 15 metrics to show the level of effect on the predicted performance level probability. The probability of a reduced lap-time increases with a metric's SHAP value. The SHAP value for each metric in the bar graph is given by the length of the bar. The absolute SHAP value shows us how much a single metric affected the prediction. All metrics are shown in the order of global metric importance, the first one being the most important and the last being the least important one. We found that the top 5 most important metrics contributing to the model were the averages of (i) speed, (ii) absolute lateral acceleration, (iii) steering reversal rate per lap, (iv) lane deviation and (v) absolute steering angle. In addition, the SHAP scatterplot in this observation shows the positive or negative impact of each metric on the predicted performance level through different colours. Each dot in the plot represents a specific feature and its associated SHAP value. The position of each dot along the x-axis shows the impact or importance of the corresponding feature on the model's predictions. More accurately, features with dots located further to the right have a higher impact on increasing the model's output, while features with dots to the left have a higher impact on decreasing the model's output. For instance, higher speed leads to a decrease in lap time, while lower levels of oversteer result in an increase in lap time.

Based on the distribution of the red and blue dots, we can gain a basic understanding of the directionality influence of metrics. Positive values indicate that they contribute to shorter lap-time, which in turn improve driving performance and vice versa. For example, high values of the average of speed have a high negative contribution on the prediction, while low values have a high positive contribution. The average of absolute steering angle has a high positive contribution when its values are high, and a low negative contribution on low values. More accurately, higher values in average of speed correspond to shorter lap-times, whereas higher average of absolute steering angle is associated with higher lap-time. The averages of maximum value of steering and throttle application have almost no contribution to the prediction, whether their values are high or low.

In addition, we investigated if the lap-time could be predicted by simply looking at the telemetry data for a particular track segment

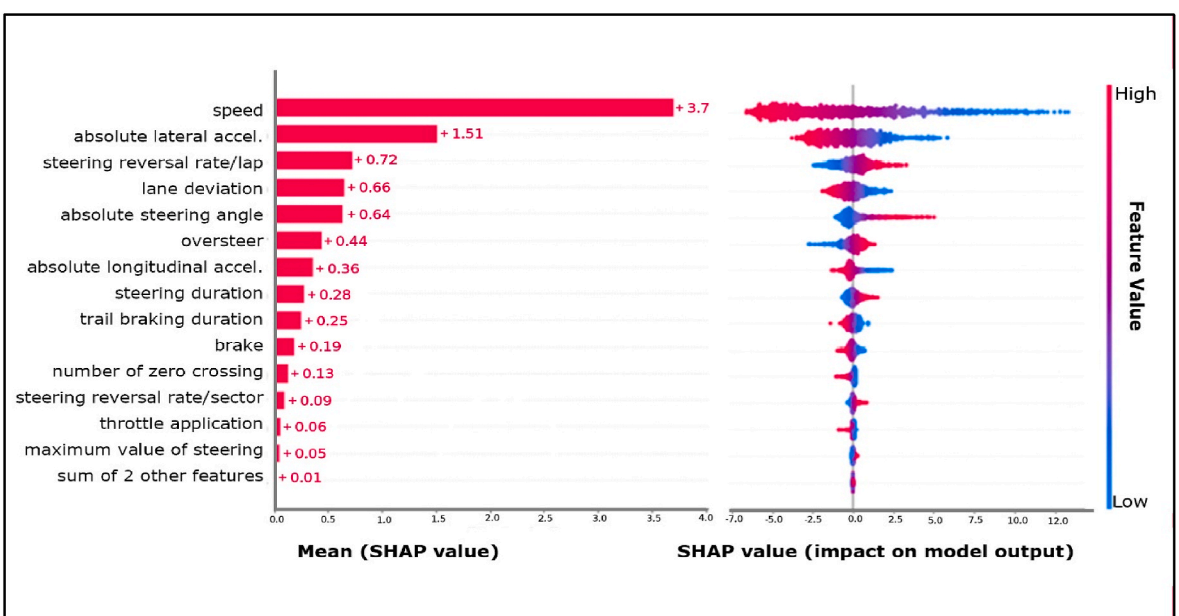


**Fig. 6.** The bar graph of the SHAP summary graph shows the effect of each metric on the XGBoost model. Speed was the factor that contributed the most to the prediction result, and absolute lateral acceleration, steering reversal rate per lap, lane deviation and absolute steering angle also had a higher contribution to the prediction result. The scatter plot represents the relationship between the metric value and the predicted probability through colour, including positive and negative prediction impacts. The greater the value (red), the greater the probability of shorter lap-time.

rather than the data for the complete lap. For instance, it would be useful to determine whether certain lap segment is crucial for predicting performance over the entire lap. Fig. 7 displays the results of this analysis and illustrates how effectively the classification models can obtain the most important metrics that have high influence on the segment time for the different types of segments (i.e., corners; straights). Considering the key performance metrics for track segments (Fig. 7), longitudinal acceleration, throttle and steering angle inputs play a significant role in influencing sim racing performance in cornering. Particularly, besides speed, longitudinal acceleration and ROTY and oversteer are three significant metrics derived from steer angle, implying turning in at the proper location of the corner to get the optimal racing line led to a shorter lap-time. These metrics seem to have some relevance to the turn-in point and apex, two key components of a racing line, which is the route a racing driver follows to take corners as quickly as feasible.

### 4.3. Driving patterns investigation of racers in SLOW and FAST lap-time clusters

To address the second aim regarding the disparities between FAST and SLOW laps (A2), we conducted time-series analysis and an investigation on the most critical metrics to obtain objective measures and offer a comparative assessment between FAST and SLOW laps. Table 2 represents the proportional difference between SLOW and FAST lap groups for each of the most important metrics. The value for each aggregated metric was derived from time-series driver dataset by computing the average of the metrics across the laps (Table 2) and corners (Table 3). The complete results of all metrics for both corners and straight segments can be found in supplemental file 9.

Fig. 8 show driving patterns for different metrics for both groups FAST (red colour) and SLOW (blue colour). The graph reveals that the beginning part of the track has the most differences in the groups. The difference is more obvious in the first three corners specially second corner (T2), which is a hairpin style corner that requires threshold

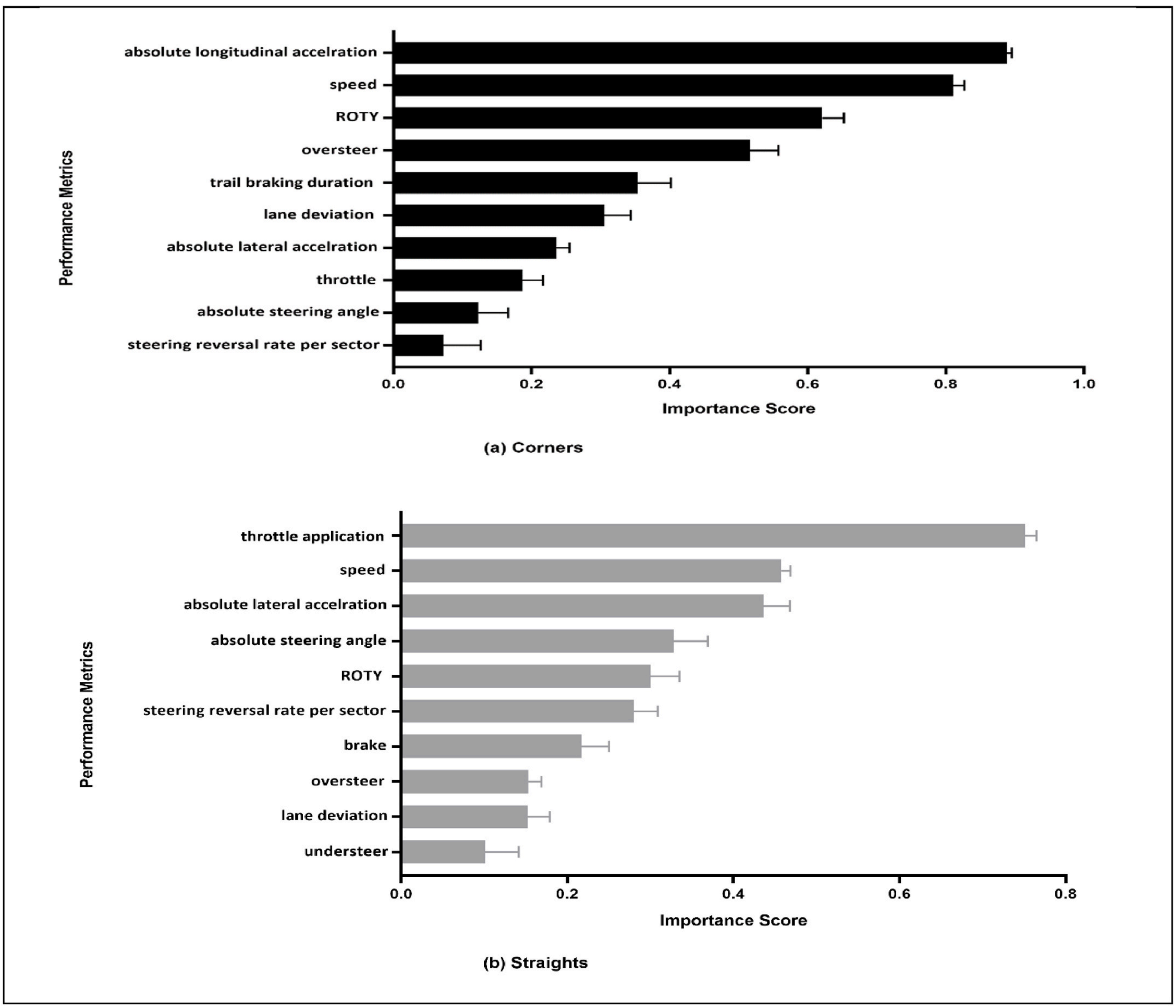


**Fig. 7.** Metrics shown to be crucial to forecast segments of lap's performance: (a) corners, (b) straights. The results in (a) were gained by running feature ranking algorithm on each corner, then averaging all importance metrics scores obtained for each corner. Error bars represent the standard deviation around each bar peak. The same approach is for straight.

**Table 2**
Proportional difference between SLOW and FAST lap groups for 15 most important metrics. The field effect shows whether the metric have positive (+) or negative (−) contribution on the prediction.

| Metric | FAST | SLOW | FAST: SLOW | Effect (±) |
|---|---|---|---|---|
| **speed** | 146.28 | 109.92 | 1.33 | – |
| **absolute lateral acceleration** | 0.72 | 0.51 | 1.4 | – |
| **steering reversal rate per lap** | 14.15 | 16.04 | 0.88 | + |
| **lane deviation** | 0.24 | 0.18 | 1.29 | – |
| **absolute Steering angle** | 28.98 | 44.43 | 0.65 | + |
| **oversteer** | 0.04 | 0.13 | 0.31 | + |
| **absolute longitudinal acceleration** | 0.42 | 0.31 | 1.35 | – |
| **steering duration** | 601.67 | 854.77 | 0.70 | + |
| **trail braking duration** | 18.28 | 13.2 | 1.38 | – |
| **brake** | 10.27 | 6.9 | 1.48 | – |
| **number of zero crossing** | 8.1 | 6.4 | 1.26 | – |
| **steering reversal rate per sector** | 0.70 | 0.81 | 0.88 | + |
| **throttle application** | 252.68 | 189.4 | 1.33 | – |
| **maximum value of steering** | 76.79 | 84.4 | 0. 9 | + |

**Table 3**
Proportional difference between SLOW and FAST lap groups during cornering. The field effect shows whether the metric have positive (+) or negative (−) contribution on the prediction.

| Metric | FAST | SLOW | FAST: SLOW | Effect (±) |
|---|---|---|---|---|
| **absolute longitudinal acceleration** | 0.41 | 0.09 | 4.39 | - |
| **speed** | 126.03 | 87.55 | 1.44 | – |
| **ROTY** | 19.41 | 15.31 | 1.26 | – |
| **oversteer** | 0.174 | 0.70 | 0.24 | + |
| **trail braking duration** | 2.46 | 6.80 | 0.36 | + |
| **lane deviation** | 0.53 | 0.28 | 1.93 | – |
| **absolute lateral acceleration** | 1.16 | 0.66 | 1.76 | – |
| **throttle** | 54.13 | 22.73 | 2.38 | – |
| **absolute steering angle** | 34.69 | 48.79 | 0.71 | + |
| **steering reversal rate per sector** | 1.42 | 1.02 | 1.38 | – |

braking in a straight line. Regardless of the big difference in the first part of the track, there is a little variation in the last segment of the track (i.e., last two corners), showing that the racers in the FAST laps got off the throttle a bit earlier, while they used roughly the same amount of brake pressure entering.

From our investigation on speed, throttle, brake, steering angle and acceleration metrics, we noticed that racers in the FAST laps accelerate earlier and more quickly after each corner, with a sharp throttle and higher brake and stable steering control. Moreover, we found that racers in the FAST laps press the throttle earlier with greater magnitude, 2.38 times more compared to SLOW laps (see Table 3) while releasing the brake earlier. Concerning the variation of peak values in brake between the FAST and SLOW groups, we employed the *t*-test method to assess the differences. The resulting p-value of 0.002 indicates a significant distinction between the FAST and SLOW groups regarding the maximum amount of braking. This significance is visually evident in the brake trace depicted in Fig. 8. Upon examining the start and end points of the brake trace's peaks in both groups, no substantial differences were observed. However, our analysis revealed a notable distinction that racers during FAST laps reach the peak 33 times more rapidly than the racers in in SLOW laps. This significant difference becomes particularly noticeable in the last five corners of the track in Fig. 8.

Looking at the trace line for both groups, we observed that in the FAST laps, racers apply the brakes right before entering the corner, then trailing off just beginning to turn. They are doing this to aid in the initial rotation of the car and to help them reach the first apex of the corner so they can turn. Thus, they keep a tighter line through this section and then manages to get on the throttle a little bit smoother and more consistently than racers in SLOW laps. As an example, looking at the third corner (T3 in Fig. 8), we see during the SLOW laps, racers get on the brakes earlier and off the throttle earlier, ended up over less speed and slowing the corner.

A significant consistency is shown when comparing driving patterns (Fig. 8) with the feature rankings shown in Figs. 6–7. The results reveal that both groups of racers follow entirely different patterns when handling corners. Pro-racers drive with a longer throttle application, more brake mean, and higher acceleration in corners. More precisely, FAST racers take full throttle on the straight considerably longer than the SLOW racers do, thus, they get on the brake a bit later, maximizing their time going at their top speed.

Upon a thorough investigation of the steering angle trace, we found that, on average, when braking values exceed 35%, racers in FAST laps maintain an average steering angle of $-6.5^{\circ}$, whereas during SLOW laps, the average steering angle is $-29.12^{\circ}$. We can also observe how quickly the steering decreases when the throttle is increased in the FAST laps. Furthermore, it appears that racers in the FAST laps apply higher amount of steering angle when approaching the corners, resulted in FAST laps having 1.93 times more lane deviation compared to SLOW laps. During FAST laps, occurrences of oversteer are 0.24 times less frequent compared to the oversteer occurrence during SLOW laps.

In order to conduct a deeper examination of the interaction between braking, acceleration, and turning, we calculated the lateral acceleration and longitudinal acceleration averages per lap distance across all laps in each group to obtain the traction circles for both FAST and SLOW laps. Then, we utilized the average values obtained for lateral and longitudinal acceleration to create an X–Y graph for each group (see Fig. 9). This graph is referred to as a *G-G plot* or *traction circle*. The shape of the graph tells how much trail braking the driver is using, or how hard he is accelerating while leaned over exiting corners. A more diamond-shaped chart would indicate fewer instances of combined accelerations than a more circular chart would. Looking at both circles in Fig. 9, it is easy to see that while racers in the FAST laps drive more around the rim of the traction circle, racers in SLOW laps follow a more direct path for a maximum braking to maximum cornering force. Racers in FAST laps often exhibit higher lateral and longitudinal forces within the circle, maximizing tire grip for sharper turns and quicker accelerations. They tend to utilize a wider range of the traction circle to achieve faster lap-times.

### 4.4. *Clustering analysis of track corner data*

Referring to the third aim of the study, which was to determine the difficulty level attributed to corners across the Brands-Hatch circuit, Fig. 10 illustrates the findings of the clustering analysis of data. The stacked bars show three groups of the corners that we have labeled difficulty level 1, difficulty level 2, and difficulty level 3. As we see, all data related to the second corner (T2) has been categorized in the same group, referring to level 3. For the second level of the corners (T3, T4, and T8), very few rows have been categorized into different groups; however, the majority of the rows are clustered together. Similarly, the rest of the corners (T1, T5, T6, T7, and T9) have been grouped in group level 1. Note that in this analysis, the clustering algorithm applied to the dataset achieved an accuracy rate of 86.34%. This accuracy shows how well the algorithm correctly groups data, matching the expected clusters in the dataset.

## 5. Discussion

The main purpose of the present study was to model and analyze race driver behaviour and determine the most important in-game metrics that dictate performance for sim racing. Specifically, we aimed to firstly identify the metrics that best predict sim racing performance, across different driver levels of expertise. To do this, we designed and tested machine learning algorithms to predict lap-time performance in sim

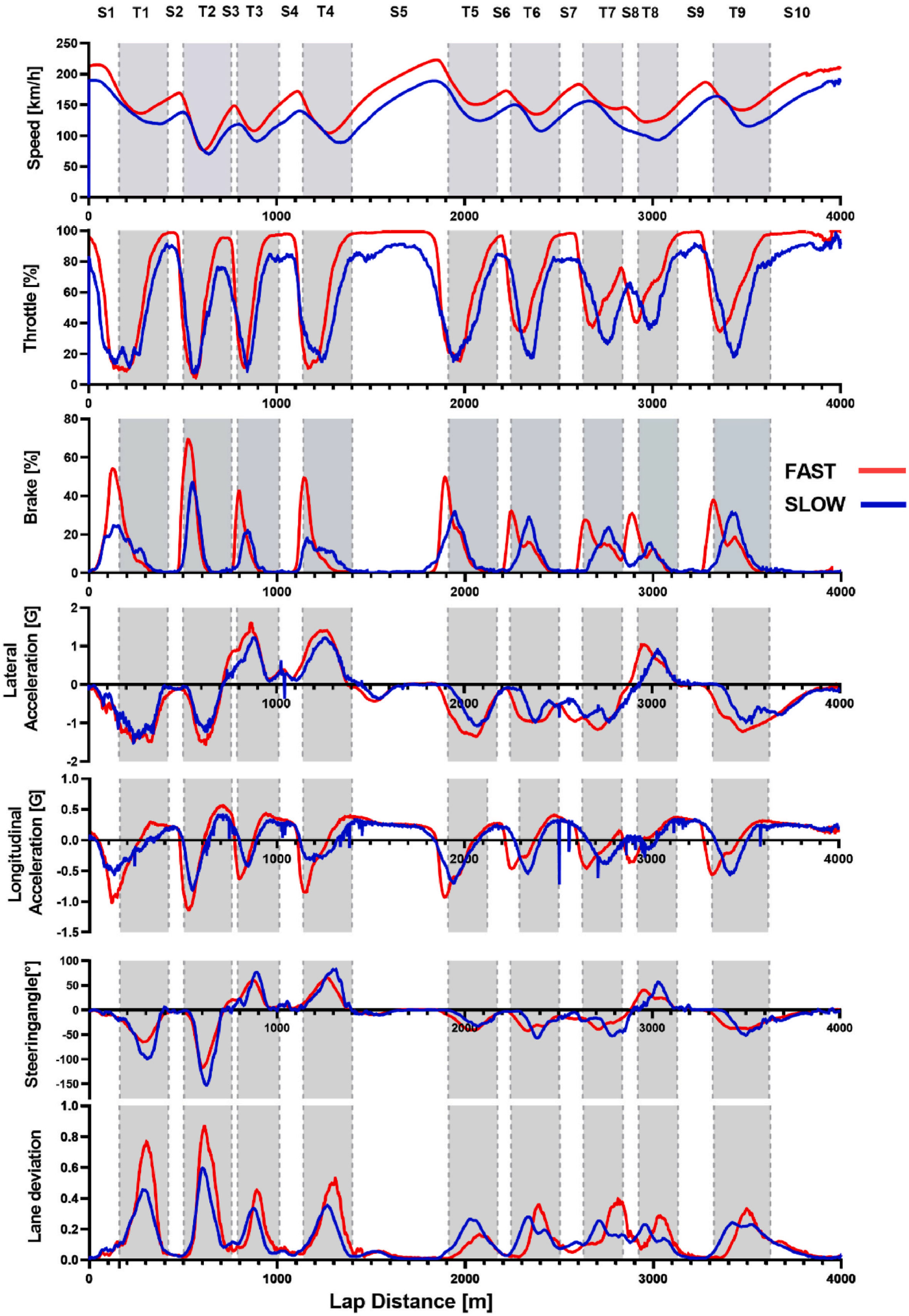


**Fig. 8.** Driving patterns for different metrics for the Brands-Hatch circuit, providing a comparison between FAST (red) and SLOW (blue) laps. The Y-axis (from top to bottom): speed, throttle position, brake position, lateral acceleration, longitudinal acceleration, steering angle, and lane deviation. The vertical dash lines in the figures depict the track segments' boundaries. Corners are specified with grey colour. The patterns show FAST laps have characteristics with a high throttle, lateral and longitudinal acceleration mean, a low percentage of trail braking duration and a low steering wheel variation.

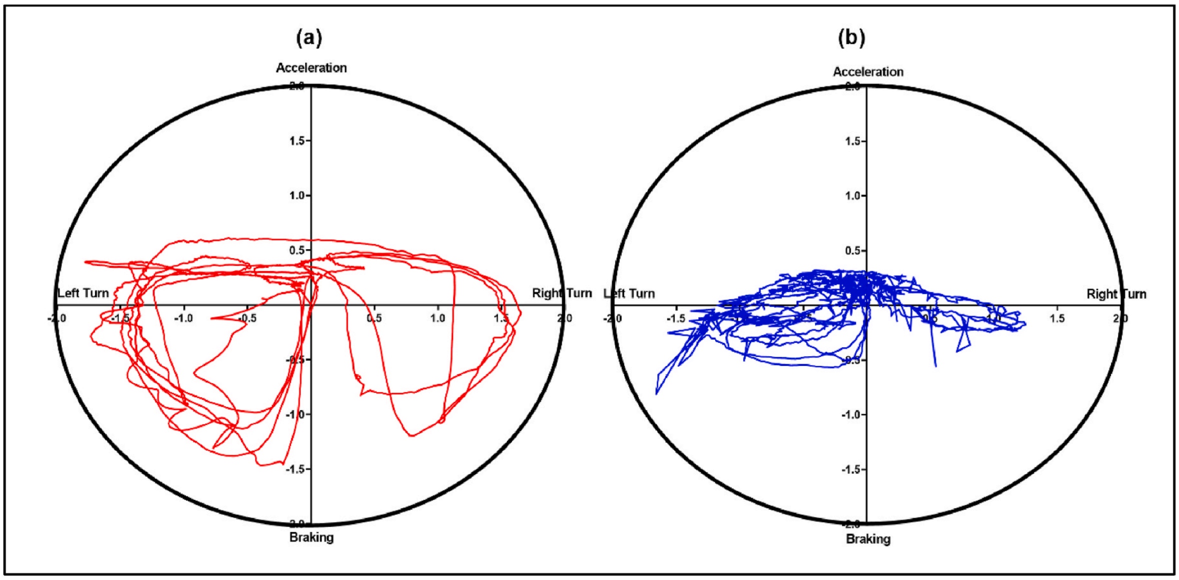


**Fig. 9.** X–Y graph of lateral (X) and longitudinal (Y) acceleration, G-G plot, for FAST (a) and SLOW (b) laps. Points in the upper half of each graph, denote acceleration, while those in the lower half denote braking (negative acceleration). Points on the right half denote turns to the right; on the left, turns to the left. Racers in FAST laps demonstrate higher lateral and longitudinal forces, having more brake and optimizing tire grip to navigate sharper turns and achieve faster accelerations.

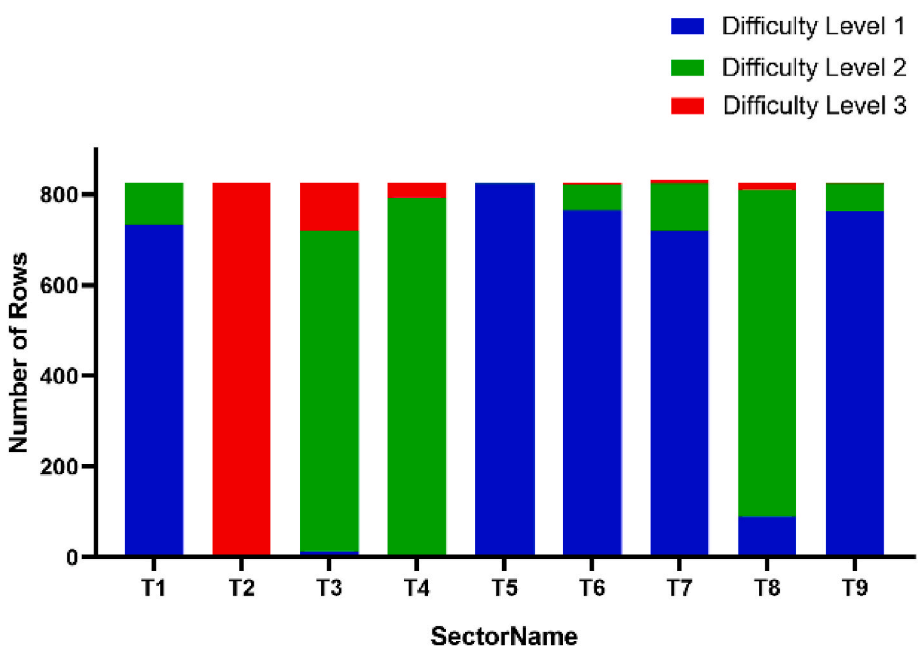


**Fig. 10.** Stack bars displaying the results of clustering of segment-specific behavioural dataset including 7847 corner rows. Difficulty level 3 includes T2, referring to second corner in the Brand hatch circuit, which is a hairpin corner. T3, T4 and T8 are categorized in difficulty level 2, and the remaining corners are classified in difficulty level 1.

racing. To determine the metrics that impact the prediction model the most, we depicted the SHAP summary plot of the model and the top 15 metrics of the prediction model (Fig. 6). The key finding from the first aim is that metrics derived from throttle, brake and steering angle play a major role in racing performance across an entire lap. More specifically, the averages of speed, lateral acceleration, steering angle, lane deviation, and steering reversal rate per lap were found to be the most important metrics within higher racing performance level.

Focusing on most important metrics obtained from our analysis, we were able to address the second aim of the study, which was to investigate the behavioural patterns between FAST and SLOW groups of laps in terms of vehicle control behaviour. We applied time-series analysis on most important metrics and obtained several metrics to investigate driving patterns and identify the key differences in driving techniques. The patterns indicate that FAST laps exhibit characteristics with high average values in throttle, lateral, and longitudinal acceleration, along with a minimal level of variability in trail braking duration. While the findings suggest increased occurrence of lane deviation in FAST laps, they also suggest reduced typical variability in steering angle.

Lastly, we conducted a track segment analysis with the goal of fulfilling the third objective (A3), aiming to ascertain the level of corner difficulty in Brands-Hatch circuit, which led to identification of three levels of corner difficulty in Brands-Hatch circuit. Such kind of analysis would also be helpful in determining corner segments. The breaking part of a corner, where the car must sufficiently slow down to prepare for the turn-in point; the racing line at a corner, which is the segment between the turn-in point; the apex point, which is the inside midpoint of the corner, and the outside apex, where the driver must gradually accelerate out of the corner, can all be identified by studying the driving styles of professional racers.

The key findings from our analysis help evaluate various aspects of performance in sim racing, such as lap-times, speed, braking efficiency, steering inputs, and consistency in driving lines. Furthermore, identifying key important metrics may help game developers or simulator designers in refining and enhancing the realism and accuracy of simulators. This contributes to creating more engaging and realistic racing experiences.

Another key finding from this study is driving patterns that is the driving patterns that aid in distinguishing the key disparities between elite and novice drivers. This provides valuable insights into what separates high-performance racers from slower ones. Understanding the techniques used by faster racers can guide slower racers in adopting better driving strategies and improving their skills. It helps in finding optimal racing lines, racing strategy development, and other techniques used by high-performance racers, leading to better race strategies.

Lastly, our findings demonstrated that driving behaviours in some track segments were much more important to the success of the entire lap than others. Consequently, the results can be used to tailor driver training programs, providing feedback and guidance to competitive sim racers for enhancing their skills in specific track segments, leading to more efficient driving practices.

## 6. Limitations and future research

Notwithstanding our findings, there are certain limitations to our study which must be acknowledged. Firstly, MoTec provides no information regarding off-tracking. Off tracking can occur at any point during a lap but primarily occurs during cornering when entire vehicle, including all its wheels, moves beyond the track boundaries. Although some of the off-tracking laps whose lap-times exceeded the outlier threshold were eliminated in the pre-processing step, a few off-tracking laps may have still been included in the final data set. A possible solution for this issue is to develop a program with ACCShared memory (Simulazioni, 2019), a library including in-game variables, that can be run concurrently with the game, reading data from the game.

Another limitation of the study is that our clustering produced SLOW cluster with a small number of laps (n = 477) compared to the number of laps included in the FAST cluster (n = 850). As the majority of participants were avid sim racers, the low frequency of low-skill racers is reflected in the overall sample. Performance of the classifier may have been impacted by the resulting class imbalance issue. Using a cross-validation technique, we established 10-fold samples with roughly identical target class distributions to overcome this problem. Classifier performance significantly improved when stratified data were used for algorithm training. Utilizing various performance metrics (confusion matrix, F1 score, and AUC) for assessing the effectiveness of classification algorithms is another strategy we employed to address the problem of the unbalanced dependent variable. Another viable approach is employing XGBoost, which supports sample weighting and can effectively address imbalances present in the dataset. We intend to explore this approach in future analyses. The choice of machine learning algorithms in this study was based on the findings of earlier research indicating that Random Forest (Smithies et al., 2021), Gradient Boosting (Semenov et al., 2017) and Support Vector Machine (Touryan et al., 2016) are effective means for forecasting game outcome particularly when the datasets are unbalanced. Consideration of additional ML approaches may potentially improve sim racing performance in the future.

In this work, we were able to conduct a comprehensive examination of driving behaviours on a particular track. Our approach is designed to be generally applicable to any track. While our current study focuses on results from a single track, the methodology and algorithms employed are not specific to that track. The developed algorithms and analysis are designed with flexibility, allowing for the analysis of telemetry data from any track, thereby demonstrating the versatility and generalizability of our methodology.

Based on the hereby generated insights, the study can be expanded by applying ML techniques for trajectory prediction and modelling of race driver behaviour. The model is useful to identify the optimal racing line in sim racing. Additionally, through an in-depth analysis of different corners, we could offer useful insights to engineers, helping them refining vehicle configurations and develop strategies to enhance performance. The knowledge from this analysis can also be transferred to self-driving vehicles to offer a technique to enhance their driving style.

## 7. Conclusion

In this work, we adopted data science solution for predicting sim racing performance utilizing telemetry data and performed a comprehensive analysis for predicting sim racing performance. Given telemetry data from 174 participants and 1327 laps, a cluster analysis was used to divide the resulting laps into SLOW and FAST groups based on the performance (lap-time). The features set was then reduced by choosing a subset of the original features of dataset by applying a hybrid wrapper-filter feature selection approach. Using this technique, three machine learning models were developed for predicting the ultimate performance throughout the course of the lap and segments. The k highest ranked metrics were then obtained from XGboost model with a maximum accuracy of 97.19% which means our model is very successful in predicting lap-time and providing the most influential metrics for racing performance. We also used the SHAP value technique to illustrate the positive or negative effects of the top 15 metrics attributed to the XGboost model. From the results, we can conclude that over the course of the whole lap, higher value of speed mean, along with higher lateral acceleration and lane deviation leads to shorter lap-time. In addition, a shorter lap-time may arise with applying greater trail braking duration and longitudinal acceleration, while reducing steering angle variation in cornering. In essence, the findings of this study may lead to a deeper understanding of driving behaviours and performance factors, ultimately contributing to enhance skills, improve training programs, develop better strategies, optimize vehicle setups, and promote competition within the sim racing community.

## CRediT authorship contribution statement

**Fazilat Hojaji:** Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, machine learning. **Adam J. Toth:** Writing – review & editing, Methodology, Conceptualization. **John M. Joyce:** Methodology, data collection. **Mark J. Campbell:** Writing – review & editing, Conceptualization.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

The main datasets have been included as supplementary files

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## Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.chbr.2024.100414.

## Appendix

**Table A1**
specification of the original datasets

| Datafile name | description | Number of rows | Number of columns |
|---|---|---|---|
| Lap_time data | The file contains a summary of all laps across all participants, providing general details of the racing session (such as venue, track name, vehicle, and duration), as well as lap times and segment times for both straights and corners. | 1327 | 26 |
| lap-specific behavioural data | The file comprises the average of driver and vehicle in-game behaviour features for each lap/participant | 1327 | 74 |
| segment-specific behavioural data | The file comprises the average of driver and vehicle in-game behaviour features for each segment/lap/participant | 15,694 | 74 |
| time-series driving data[a] | The file contains time-series data representing the driver's behaviour features. | 22,800 | 87 |

[a] There is one file available for each participant, resulting in a total of 174 files for all participants combined.

**Table A2**
Description of performance metrics requiring detailed descriptions.

| Feature Name | Description | unit | MoTec Data Feature (C)/Math (M) |
|---|---|---|---|
| Lap-time | The amount of time it takes for a vehicle to complete one lap of a track. | s | C |
| Segment time | The amount of time it takes for a vehicle to complete one segment (straight or corner) of a track. | s | C |
| Lap distance | the length of one complete track or lap | m | |
| Speed | The rate at which the vehicle is moving at any given time on track | km/h | C |
| RPMS | the engine's rotations per minute | 1/min | C |
| Steering angle | The amplitude applied by the driver to the steering wheel at any given time on track. | degree | C |
| ROTY | the angle of wheel or the direction of the wheel in right or left | degrees/s | C |
| Lateral acceleration | the level of acceleration of the car in lateral direction (side to side) | $m/s^2$ | C |
| Longitudinal acceleration | the level of acceleration of the car in a longitudinal direction (forward and back) | $m/s^2$ | C |
| Brake | The amplitude applied by the driver to the brakes at any given time on track. | % | C |
| Braking duration | the time from where the driver presses the brake pedal to the moment the driver releases the brake pedal. | s | M |
| Trail braking duration | the period during which a driver applies the technique of trail braking while navigating a corner | s | M |
| Throttle | The amplitude applied by the driver to the throttle | % | C |
| Throttle application | the time from where the driver presses the gas pedal to the moment the driver releases the gas pedal. | s | M |
| Maximum value of steering | the moment when the driver applies the greatest steering input, usually during a turn or maneuver | degree | M |
| Cumulative angular travel | The total rotation or angular movement of steering wheel over a given lap | degree | M |
| Angular velocity | measures how quickly steering wheel is rotating or changing its angular position as time passes. | degree/s | M |
| Total number of direction change | the count of times the direction of the steering wheel is changed during driving (Each time the driver turns the steering wheel in a different direction, it constitutes a direction change) | absolute value | M |
| Lane deviation | The value of the difference between the car's lateral position and the centre of the lane | degree | M |
| understeer | number of times during a lap, the front tyres generate a greater slip angle than the rear tyres | absolute value | M |
| oversteer | Number of times during a lap, the rear slip angle exceeds the front slip angle | absolute value | M |
| Steering reversal rate per lap | The frequency of lane deviations occurring within a single lap | absolute value | M |
| Steering reversal rate per segment | The frequency of lane deviations occurring within a single lap segment | absolute value | M |
| Number of zero crossing | the count of times steering angle crosses the zero-axis or baseline in a given lap | absolute value | M |

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